

Arash Mousavi

INFO-I416-24528

Mentor: Prof. Ghansham Manwani

12/13/2025

Introduction

Cloud computing has fundamentally changed the way advanced computing is developed, deployed, and accessed. Rather than requiring individuals or businesses to own specialized hardware and implement the complex algorithms from scratch, the cloud platforms provide robust and powerful computational resources and the artificial intelligence (AI) capabilities as on-demand services. This change has democratized the access to AI tools and enables sentiment analysis, image recognition, and natural language processing (NLP) being performed through simple application programming interfaces (APIs). Other than the accessibility, the underlying mathematical foundations of these systems are usually hidden from most of the users and this create a gap between the practical usage and theoretical understanding.

Linear algebra is the backbone of the modern AI systems. Concepts like vectors, matrices, dot product, and similarity measures are the mathematical keywords and concepts which data is represented and processed. Neural networks, embedding models, and optimization algorithms rely on linear transformation. Cloud-based AI services produce outputs that reflect these same mathematical principles. Understanding this association and connection is essential for interpreting cloud AI behavior, evaluating model reliability, and fostering transparency and ethics in AI.

In addition to the classical AI techniques, cloud computing is becoming the main access point for emerging the computational paradigms, especially quantum computing. Quantum computing is based on a fundamentally different mathematical framework and is based on complex vector spaces, unitary transformations, and probabilistic measurements. Quantum computing is still in its early stages, but it is already being offered through cloud platforms as experimental services, positioning the cloud as a bridge between classical and quantum computation. The cloud computing is not only a platform for deploying today's AI models but it is also a testing ground for future technologies that may change how computations are performed.

In this Honors Project research, we examine cloud computing through a mathematical perspective while focusing on both present and emerging technologies. The first part of the paper has a focus on the role of linear algebra in cloud-based AI models by analyzing and offering a combination of theoretical explanation and empirical analysis. A simple, interpretable linear algebra baseline is built and compared with sentiment analysis outputs from cloud AI services, and it highlights the alignment between the mathematical foundation and real-world platforms.

By combining cloud computing, linear algebra, and quantum computing within a unified framework, this project shows how mathematical foundations continue to form both current capabilities and future pathways of the cloud computing. *

Linear Algebra Fundamental Applications In Cloud AI

Vector Representation of Language

Many AI applications, especially sentiment analysis, require raw textual data to be converted into numerical form (Arrays) before computation can take place. A group of eight English sentences representing positive, negative, neutral, and ambiguous sentiment was chosen. Each sentence was converted into a bag-of-words vector and got normalized leveraging the Euclidean norm.

A sentiment polarity axis was defined by assigning positive weight to words linked and associated with positive sentiment and also negative weights to words associated with negative sentiment. Each sentence vector was projected onto this axis using a dot product (the resulting values were normalized to lie within the range of $[-1,1]$). This process gave up a scalar polarity score for each of the sentences and that served as an illustrative baseline for sentiment classification.

It is worth mentioning that this baseline is not intended to replicate the production-level AI models, and it only serves as an interpretable mathematical tool for showing how linear algebra can capture the basic sentiment structure.

Dot Product, Norms, & Cosine Similarity

Cosine similarity is one of the most widely used similarity measures in the vector-space models. It measures the angle between two vectors, rather than their absolute values. This makes it useful for comparing textual data, as it considers direction rather than length.

The cosine similarity formula:

$$\text{Cosine Similarity (x,y)} = \frac{x \cdot y}{|x||y|}$$

In the context of sentiment analysis, cosine similarity can be used to represent sentence vectors onto a predefined sentiment direction, yielding a scalar value that shows polarity. The positive values indicate alignment with positive sentiment, and the negative values indicate alignment with negative sentiment. Also, the values near zero indicate neutrality.

Neural Networks as Linear Algebraic Systems

The modern AI models are composed of layers that perform linear algebraic operations. Each neural network layer calculates an affine transformation of its inputs and it hand the result to a nonlinear activation function:

$$y = f(Wx + b)$$

W is a weight matrix, x is an input vector (array), and b is a bias vector, where f is a nonlinear function (like Sigmoid).

Cloud AI Services

We evaluated two well-known and commercial AI services in this study:

- Amazon Comprehend: This is a sentiment tool from AWS, and it provides sentiment labels with confidence scores for positive, negative, and mixed categories.
- Microsoft Azure Foundry: This also provides sentiment labels and confidence scores for positive, neutral, and negative sentiment.

Identical eight sentences were input to both of the services, and JSON outputs were parsed, and the sentiment confidence scores were normalized.

Result & Analysis

Cosine Similarity

The cosine similarity matrix computed from the bag-of-words vectors demonstrates clear structural patterns. Sentences that share sentiment-related vocabulary show moderate similarities, and neutral or ambiguous sentences show low similarities to the strong polarized examples.

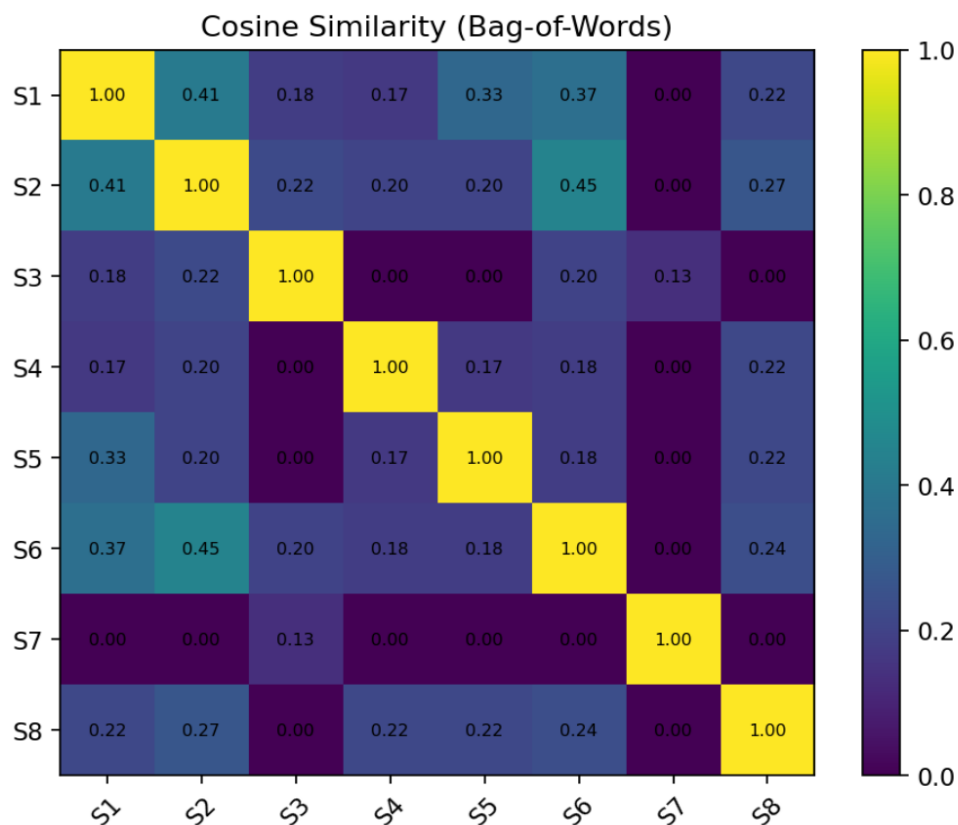


Figure 1: Cosine Similarity matrix for sentence vectors using a bag-of-words representation.

This visualization indicates that linear algebraic similarity measures catch the lexical relationships but struggle to encode the nuanced sentiment in ambiguous cases.

Comparison with Cloud AI Outputs

Normalized sentiment scores from the Amazon Comprehend and Azure Foundry were compared to the baseline polarity values. Scatter plots illustrate the relationship between the mathematical baseline and cloud AI outputs.

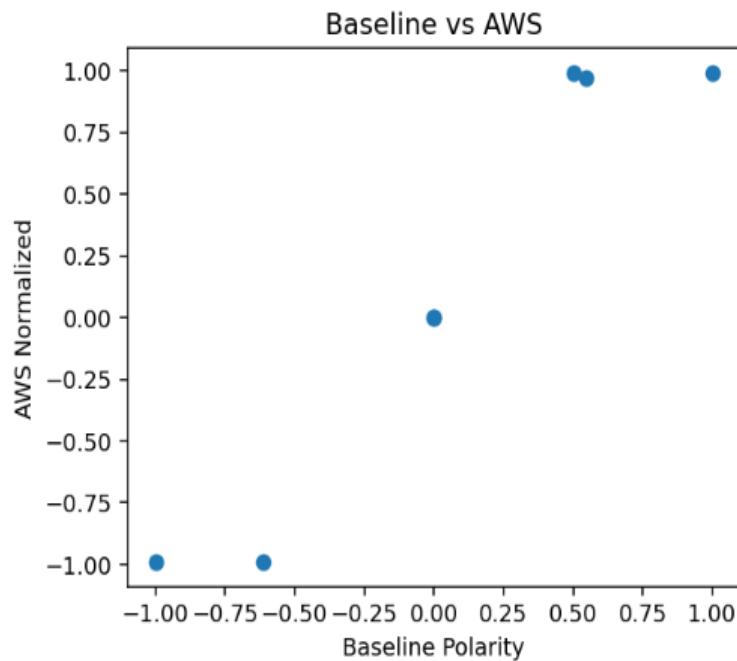


Figure 2: The relationship between baseline polarity scores and the Amazon Comprehend normalized sentiment outputs.

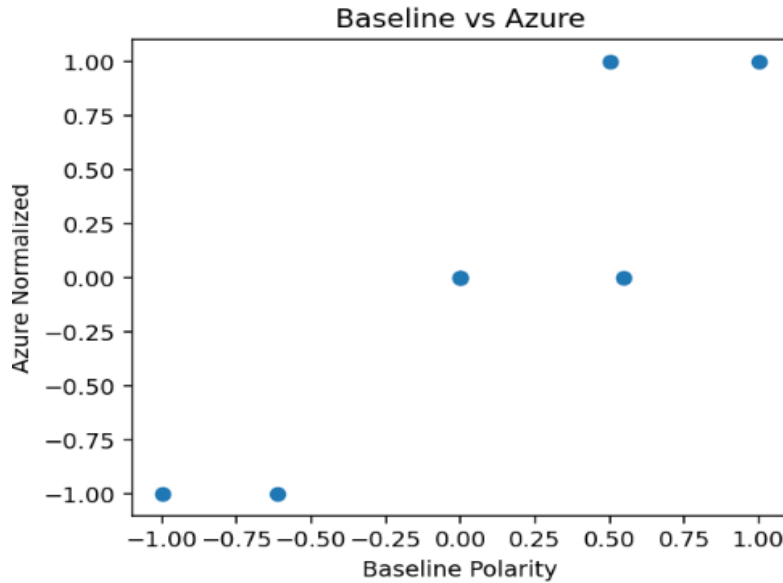


Figure 3: The relationship between baseline polarity scores and the Azure Foundry normalized sentiment outputs.

Both of the platforms show robust alignment with the baseline for positive and negative sentences, but Azure tends to assign neutral classifications more for the borderline cases. Also, AWS produces more extreme polarity values. This indicates some differences in model calibration.

Quantitative Correlation Analysis

The correlation analysis even more supports the observations. The baseline polarity scores indicate a strong positive correlation with both Azure and AWS normalized outputs (correlation coefficient exceeds 0.9). Also, AWS and Azure outputs are strongly correlated with each other.

	baseline_polarity	aws_norm	azure_norm
baseline_polarity	1.000000	0.959544	0.913618
aws_norm	0.959544	1.000000	0.909091
azure_norm	0.913618	0.909091	1.000000

Table 1: The correlation matrix comparing baseline polarity, AWS normalized scores, and the Azure normalized scores.

These results highlight that despite the complexity of the cloud AI models, their sentiment outputs still remain aligned with the linear algebraic representation of the language.

Discussion & Implications for Cloud Computing

The results show that cloud-based sentiment analysis services preserve fundamental linear semantics structures. The modern AI models leverage deep architectures and contextual embeddings, and their outputs still reflect the relationships that can be approximated leveraging simple vector-space methods.

Our findings have significant implications for cloud computing. At first, it suggests that cloud AI platforms are not mathematically arbitrary, and then it highlights the value of interpretable baselines for validating and understanding systems. At the end, it raises ethical considerations related to transparency and accessibility. By leveraging foundational mathematical principles, users can examine and evaluate the cloud AI outputs without having technical access to the internal models.

Our analysis establishes a conceptual and mathematical foundation for the rest of the paper. Quantum computing introduces new mathematical structures and computational possibilities.

Quantum Computing

The classical computers show the information using bits that take on one of two discrete values of either 0 or 1. All the classical algorithms, such as those used in cloud computing, operate through sequences of binary states. Quantum computing, on the other hand, introduces a fundamentally different unit of information known as the quantum bit or qubit. Unlike the classical binary bits, qubits can exist in a superposition of state, which means they can represent a combination of 0 and 1 at the same time.

In mathematics, a qubit is shown as a vector in a two-dimensional complex vector space.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

α and β are complex numbers, and they satisfy $|\alpha|^2 + |\beta|^2 = 1$. The measurement collapses the qubit into either state $|0\rangle$ or $|1\rangle$ with probability determined by these coefficients. This probabilistic nature makes the quantum computation different from the deterministic classical computation and introduces new computational possibilities.

Linear Algebra in Quantum Mechanics

Quantum computing is conceptually different from classical computing; however, it remains rooted in linear algebra. Quantum states are vectors, quantum gates are matrices, and the quantum algorithms consist of sequences of linear transformations. Also, the quantum operations must be unitary, meaning that they preserve vector norms and are reversible.

Multi-qubit systems that use tensor products cause an exponential growth in state space. For instance, a system of n qubits is represented by a vector in a 2^n -dimensional complex space. This exponential representation underlies both the promise and challenges of quantum computing, which can promise powerful parallelism, but it also makes simulation on classical hardware very difficult.

Reliance on linear algebra provides a natural conceptual bridge between the classical AI systems discussed and quantum computing. In both types, computation is expressed by doing operations on matrices and vectors, but the constraints and interpretations are significantly different.

Quantum Algorithms & Computational Advantages

Quantum computing got widespread attention with development of algorithms that outperform the classical counterparts for specific problems. Shor's algorithm shows that integer factorization could be performed in polynomial time on a quantum computer, which poses a potential threat to classical cryptographic systems. Grover's algorithm demonstrates that the unstructured search problems could be solved in $O(\sqrt{N})$ time, which is a quadratic speedup in comparison to the classical search.

These algorithms highlight a key principle that quantum advantage is problem-specific. Quantum computers do not outperform classical computers in everything.

Implications for Machine Learning and Optimization

A lot of machine learning and optimization problems can be framed in terms of linear algebra and are potential candidates for quantum acceleration. Quantum algorithms like the Harrow-Hassidim-Lloyd (HHL) algorithm aim to solve systems of linear equations faster than the classical algorithms under certain conditions. Solving linear equations is central to optimization, regression, and kernel methods and quantum computing has been a hot topic as a possible accelerator for AI.

The practical quantum advantage in machine learning, however, remains theoretical. There are issues, including data loading, noise, and algorithm constraints. As a result, quantum computing is mostly viewed as a long-term complement to classical AI rather than an early replacement. In fact, quantum computing can provide significant gains for some specific tasks involving factorization, search, and structured linear algebraic problems.

Why Quantum Computing is Cloud-Native

Unlike the classical computing resources, quantum hardware is very expensive, fragile, and very sensitive to the environment disturbances. Keeping the quantum coherence requires extreme conditions including near-zero temperatures and electromagnetic isolation. These difficult conditions make local ownership of the quantum hardware impractical.

Additionally, quantum computing has emerged as a cloud-native technology. Cloud platforms provide remote access to the quantum processors, simulators that allow end-users to try and experiment with quantum algorithms without direct interaction with the physical hardware. This is very similar to the adoption of classical cloud computing.

Cloud Quantum Platforms

There are several major providers offering quantum computing services:

- AWS Braket: Provides access to quantum hardware along with classical simulators.
- Azure Quantum: Integrates quantum development tools with the classical cloud workflows.
- IBM Quantum: Offers cloud-based access to superconducting quantum processors. It also offers educational resources.

These providers emphasize hybrid computing, where classical preprocessing and post-processing are combined with quantum. This method reinforces the role of cloud computing as an integral integration layer.

Challenges, Limitations, and Ethical Considerations

There are some serious technical and conceptual challenges that quantum computing faces. Current quantum devices are classified as Noisy Intermediate-Scale Quantum (NISQ) systems, which means they suffer from limited qubit counts and high error rates. Error correction is an open research problem, and a large-scale fault-tolerant quantum computer is still several years away from us.

From a cloud perspective, quantum computing raises ethical and security challenges as well. The potential impacts on cryptography, unequal access to the advanced computational resources, and increased dependence on large cloud providers must be considered and anticipated thoughtfully.

Future Outlook

In the near future, quantum computing will most likely function as a computational accelerator accessed through cloud platforms. The hybrid classical-quantum workflows will spread all over the various sectors, and the quantum component will be handling specific, defined subproblems. Over the longer term, enhancements in hardware stability and error correction may improve and expand the scope of quantum advantage even further.

Also, cloud computing will be the basis of this evolution. Understanding the mathematical foundations of both classical and quantum computing is vital for engaging with the future of cloud computing.

Conclusion

This paper examined and evaluated cloud computing from the perspective of mathematics, and it connected cloud-based AI platforms with the emerging quantum technologies. The first part highlighted that cloud-based AI systems remain grounded in linear algebraic principles as shown through the experiments, and the second part explored quantum computing as a fundamentally different system but yet a mathematically related system that is getting more integrated into the cloud computing platforms.

These analyses highlight that cloud computing is not only an infrastructure layer, but it is also a mathematical and conceptual bridge between the classical and future (hybrid quantum) computation.

References

Amazon Web Services. (2025). Amazon Comprehend | <https://docs.aws.amazon.com/comprehend>

Google Cloud. (2025). *Cloud Natural Language API Documentation. | <https://cloud.google.com/natural-language/docs>

Microsoft Azure. (2025). *Azure AI Language – Text Analytics Documentation. | <https://learn.microsoft.com/en-us/azure/ai-services/language-service>

Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. <https://www.deeplearningbook.org>

National Institute of Standards and Technology (NIST). (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). U.S. Department of Commerce. | <https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf>

Jobin, A., Ienca, M., & Vayena, E. (2019). *The global landscape of AI ethics guidelines. Nature Machine Intelligence, 1*(9), 389–399. <https://doi.org/10.1038/s42256-019-0088-2>

European Union. (2024). EU Artificial Intelligence Act (AI Act). Official Journal of the European Union. <https://artificialintelligenceact.eu/>

Octaria. (2024). Amazon Web Services vs. Azure vs. Google Cloud Platform: AI Services and Products. | <https://www.octaria.com/blog/amazon-web-services-vs-azure-vs-google-cloud-platform-ai-services-and-products>

Google Cloud. (2024). AWS, Azure, and Google Cloud Service Comparison. | <https://cloud.google.com/docs/get-started/aws-azure-gcp-service-comparison>

Wiz.io. (2024). NIST AI Risk Management Framework Explained. | <https://www.wiz.io/academy/nist-ai-risk-management-framework>

Palo Alto Networks. (2024). Understanding the NIST AI Risk Management Framework. <https://www.paloaltonetworks.ca/cyberpedia/nist-ai-risk-management-framework>

Preskill, J. (2018). Quantum computing in the NISQ era and beyond. *Quantum*, 2, 79.
<https://doi.org/10.22331/q-2018-08-06-79>

Nielsen, M. A., & Chuang, I. L. (2010). *Quantum computation and quantum information* (10th anniversary ed.). Cambridge University Press.

Shor, P. W. (1997). Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Journal on Computing*, 26(5), 1484–1509.

Grover, L. K. (1996). A fast quantum mechanical algorithm for database search. *Proceedings of the 28th Annual ACM Symposium on Theory of Computing (STOC)*.

Harrow, A. W., Hassidim, A., & Lloyd, S. (2009). Quantum algorithm for linear systems of equations. *Physical Review Letters*, 103(15).

Ciliberto, C., et al. (2018). Quantum machine learning: a classical perspective. *Proceedings of the Royal Society A*, 474(2209).

McClean, J. R., et al. (2016). The theory of variational hybrid quantum-classical algorithms. *New Journal of Physics*, 18(2).

National Academies of Sciences, Engineering, and Medicine. (2019). *Quantum computing: Progress and prospects*. National Academies Press.